

LINUX FILESYSTEM

Srinivas P Bodke
Reg No: 261100690018

Linux Filesystem Investigation Assignment

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

- The storage devices available.
- The partitions available on each device.
- The filesystem type used by each partition.
- The device/partition on which `/` is located.
- The filesystem type used for `/`.
- The total, used, and available storage space.

Suggested commands

```
lsblk  
lsblk -f  
df -T  
df -h
```

Answers

Question	Command	Output / Observation	Explanation / Answer
1. What storage devices are available?	lsblk	sda — 20G — disk; sr0 — 1024M — rom	Observe `NAME`, `SIZE`, and `TYPE`. The main storage device is `/dev/sda` (20G).
2. What partitions are available on each device?	lsblk	sda1 — 18.9G — part; sda2 — 1K — part; sda5 — 1.1G — part	Entries with `TYPE = part` under `sda` are the partitions of `/dev/sda`.
3. What filesystem type is used by each partition?	lsblk -f	/dev/sda1 → ext4; /dev/sda5 → swap; /dev/sda2 → no filesystem	The `FSTYPE` column shows the filesystem type.
4. On which device/partition is `/` located?	df -T	/dev/sda1 ext4 ... /	The row containing `/` identifies the root partition. Answer `/dev/sda1`.
5. What filesystem type is used for `/`?	df -T	/dev/sda1 ext4 ... /	The `Type` column shows `ext4`.
6. What is the total, used, and available storage space?	df -h	/dev/sda1 19G 15G 2.7G 85% /	Total = 19G, Used = 15G, Available = 2.7G.

Storage Configuration Summary

Device/Partition	Filesystem Type	Size	Mount Point
/dev/sda	—	20G	—
/dev/sda1	ext4	18.9G	/
/dev/sda2	—	1K	—
/dev/sda5	swap	1.1G	[SWAP]
/dev/sr0	—	1024M	—

```
(sri@kali)-[~]
```

```
$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	20G	0	disk	
└sda1	8:1	0	18.9G	0	part	/
└sda2	8:2	0	1K	0	part	
└sda5	8:5	0	1.1G	0	part	[SWAP]
sr0	11:0	1	4.5G	0	rom	

```
(sri@kali)-[~]
```

```
$ lsblk -f
```

NAME	FSTYPE	FSVER	LABEL	UUID	FSAVAIL	FSUSE%	MOUNTPOINTS
sda							
└sda1	ext4	1.0		c7bfbcc0-8445-4bc7-bad0-2bbbf3fd7d91	2.9G	79%	/
└sda2							
└sda5	swap	1		71e66f25-ca29-48dd-b746-2137e6095968			[SWAP]
sr0	iso9660	Joliet Extension	Kali Linux amd64 1	2026-06-16-13-03-54-00			

```
(sri@kali)-[~]
```

```
$ df -T
```

Filesystem	Type	1K-blocks	Used	Available	Use%	Mounted on
udev	devtmpfs	954372	0	954372	0%	/dev
tmpfs	tmpfs	197972	1196	196776	1%	/run
/dev/sda1	ext4	19353424	15254808	3090184	84%	/
tmpfs	tmpfs	989844	4	989840	1%	/dev/shm
none	tmpfs	1024	0	1024	0%	/run/credentials/systemd-journald.service
tmpfs	tmpfs	989848	8	989840	1%	/tmp
none	tmpfs	1024	0	1024	0%	/run/credentials/getty@tty1.service
tmpfs	tmpfs	197968	116	197852	1%	/run/user/1000

```
(sri@kali)-[~]
```

```
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	933M	0	933M	0%	/dev
tmpfs	194M	1.2M	193M	1%	/run
/dev/sda1	19G	15G	3.0G	84%	/
tmpfs	967M	4.0K	967M	1%	/dev/shm
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	967M	20K	967M	1%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	194M	116K	194M	1%	/run/user/1000

```
(sri@kali)-[~]
```

```
$ df -h /
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	19G	15G	3.0G	84%	/

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

Based on the values observed from your Kali Linux system:

```
/dev/sda - 20G Disk
├── /dev/sda1 - 18.9G
│   ├── Filesystem: ext4
│   └── Mount Point: /
├── /dev/sda2 - 1K
│   └── No filesystem / mount point
└── /dev/sda5 - 1.1G
    ├── Filesystem: swap
    └── Mount Point: [SWAP]
```

`/dev/sr0` is a separate **1024 MB ROM/CD/DVD device**, so it is not part of the `/dev/sda` disk hierarchy.

Explain in 2–3 sentences the relationship between these four components.

A disk is the main storage device, which is divided into partitions. Each partition can contain a filesystem, and the filesystem is attached to a mount point so that Linux can access and organize the stored files.

For your system, you can write:

`/dev/sda` is the 20G disk that contains partitions such as `/dev/sda1` and `/dev/sda5`. `/dev/sda1` uses the `ext4` filesystem and is mounted at `/`, which is the main root filesystem of your Kali Linux system.

```
(sri@kali)-[~]
└─$ pwd
/home/sri

(sri@kali)-[~]
└─$ cd /

(sri@kali)-[/]
└─$ ls /
bin  dev  home  initrd.img.old  lib32  lost+found  mnt  proc  run  srv  tmp  var  vmlinuz.old
boot  etc  initrd.img  lib  lib64  media  opt  root  sbin  sys  usr  vmlinuz

(sri@kali)-[/]
└─$ ls -d /*
/bin/  /dev/  /home/  /lib32/  /lost+found/  /mnt/  /proc/  /run/  /srv/  /tmp/  /var/
/boot/  /etc/  /lib/  /lib64/  /media/  /opt/  /root/  /sbin/  /sys/  /usr/

(sri@kali)-[/]
└─$ ls -la /
total 76
drwxr-xr-x 19 root root 4096 Sep  3 18:46 .
drwxr-xr-x 19 root root 4096 Sep  3 18:46 ..
lrwxrwxrwx  1 root root    7 Sep  3 17:26 bin → usr/bin
drwxr-xr-x  3 root root 4096 Sep  3 18:55 boot
drwx----- 2 root root 4096 Sep  3 18:46 .cache
drwxr-xr-x 18 root root 3280 Sep 11 23:45 dev
drwxr-xr-x 186 root root 12288 Sep 13 21:33 etc
drwxr-xr-x  3 root root 4096 Sep  3 18:52 home
lrwxrwxrwx  1 root root   34 Sep  3 17:27 initrd.img → boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx  1 root root   34 Sep  3 17:27 initrd.img.old → boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx  1 root root    7 Sep  3 17:26 lib → usr/lib
lrwxrwxrwx  1 root root    9 Sep  3 18:37 lib32 → usr/lib32
lrwxrwxrwx  1 root root    9 Sep  3 17:26 lib64 → usr/lib64
drwx----- 2 root root 16384 Sep  3 17:25 lost+found
```

Exercise 3 – Explore the Root Filesystem

Start from your current location.

Question	Command	Output / Observation	Explanation
1. Determine your current working directory.	pwd	/home/Sri/linux/theory/t3	Shows the directory where you are currently working.
2. Navigate to the root directory.	cd / pwd	No output from `cd`; `pwd` should show `/`.	Confirms that you are in the root directory.
3. Display the contents of the root directory.	ls /	bin boot dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var	Shows names directly under `/`.
4. Identify all directories directly available under `/`.	ls -d /*/	/bin/ /boot/ /dev/ /etc/ /home/ /lib/ /media/ /mnt/ /opt/ /proc/ /root/ /run/ /sbin/ /srv/ /sys/ /tmp/ /usr/ /var/	Entries ending with `/` are directories directly under `/`.
5. Determine whether hidden entries are present.	ls -la /	Listing includes `.` and `..`, and any other entries beginning with `.` if present.	`-a` displays hidden entries.

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations: `/home` `/root` `/etc` `/var` `/var/log` `/tmp` `/usr` `/boot` `/dev` `/proc`

For each location:

- Examine its contents.
- Identify at least three entries, where available.
- Determine the likely purpose of the directory.

Directory	Operation	Observed Contents	Purpose
/home	ls /home	Sri	Stores personal files and directories of normal users.
/root	sudo ls -l /root	total 0 — empty	Home directory of the root user.
/etc	ls /etc	apt, ssh, systemd	Contains system-wide configuration files and directories.
/var	ls /var	cache, lib, log	Stores variable data such as logs, caches, and application data.
/var/log	ls /var/log	auth.log, syslog, journal	Stores system and application log information.
/tmp	ls /tmp	Temporary files and directories	Stores temporary data created by users and applications.
/usr	ls /usr	bin, lib, share	Contains user programs, libraries, and shared resources.
/boot	ls /boot	grub, vmlinuz, initrd.img	Contains files required to boot the Linux system.
/dev	ls /dev	null, random, tty	Contains special device files used to access hardware and virtual devices.
/proc	ls /proc	1, cpuinfo, meminfo	Virtual filesystem providing information about running processes and the Linux kernel.

```
(sri@kali)-[/]
└─$ ls /home
sri

(sri@kali)-[/]
└─$ ls /root
ls: cannot open directory '/root': Permission denied

(sri@kali)-[/]
└─$ sudo ls /root
[sudo] password for sri:
ls: cannot open directory '/root': Permission denied

(sri@kali)-[/]
└─$ sudo ls -l /root
total 0

(sri@kali)-[/]
└─$ ls /etc
adduser.conf      deluser.conf      hosts              logrotate.conf    openvpn            rmt                sysctl.d
alternatives      depmod.d           hosts.allow        logrotate.d        opt                rpc                sysstat
anacron            dhcpcd.conf        hosts.deny         machchanger        os-release         runit              systemd
aparmor            dictionaries-common ifplugd            machine-id         pam.conf           samba              terminfo
apt                doc-base           ImageMagick-7     magic              paperspecs         sane.d             texmf
arp-scan           dpkg               inetsim            magic.mime          manpath.config     scalpel            theHarvester
avahi              e2scrub.conf       initt.d            manpath.config     matplotlibrc       sddm.conf.d        tightvncserver.conf
bash.bashrc       eac                initramfs-tools    mime.types          perl                searchsploit_rc    timidity
bash_completion   emacs              inputrc            minicom             perl                security            tmpfiles.d
bash_completion.d environment         inserv.conf.d     miredo              plymouth            selinux            tmp2-tss
                                                           ts.conf
```


Answers

1. How does Linux identify a hidden file?

A file or directory is hidden when its name begins with `.`.

2. Why does a normal `ls` not display these files?

`ls` hides entries beginning with `.` by default.

3. What do `.` and `..` mean?

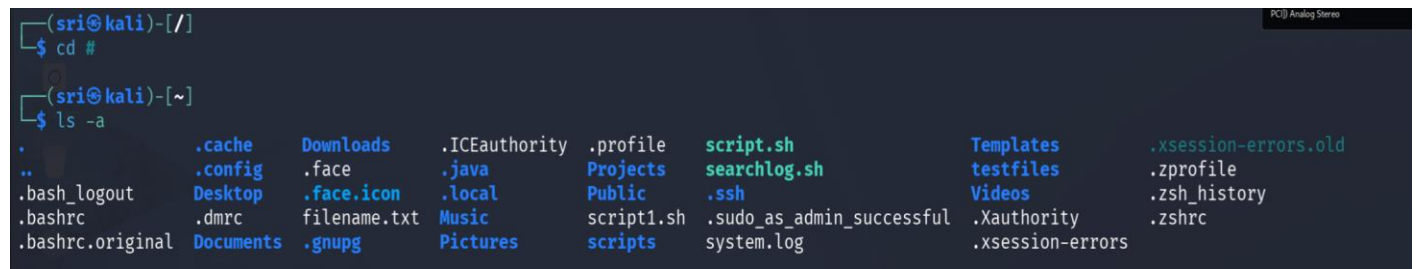
`.` means the current directory, while `..` means the parent directory.

4. Can a hidden file still be accessed if its name is known?

Yes. For example, `.bashrc` can be accessed using `cat ~/.bashrc`.

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden files can be legitimate configuration files such as `.bashrc` and `.zshrc`. However, malicious files can also be hidden, so their name alone does not determine whether they are safe or harmful.



```
(sri@kali)-[/]  
$ cd #  
  
(sri@kali)-[~]  
$ ls -a  
.  
..  
.bash_logout  
.bashrc  
.bashrc.original  
.cache  
.config  
.Desktop  
.dmrc  
.Documents  
Downloads  
.face  
.face.icon  
filename.txt  
.gnupg  
Pictures  
.ICEauthority  
.java  
.local  
Music  
Pictures  
.profile  
.Projects  
.Public  
script1.sh  
scripts  
.script.sh  
.searchlog.sh  
.ssh  
.sudo_as_admin_successful  
system.log  
.Templates  
.testfiles  
.Videos  
.Xauthority  
.xsession-errors  
.xsession-errors.old  
.zprofile  
.zsh_history  
.zshrc
```

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: `mkdir`

Hint: you can create this hierarchy in a single command

```
cyber_lab/  
|  
+-- evidence/  
|   +-- network/  
|   +-- system/  
|  
+-- logs/  
|  
+-- scripts/  
|  
+-- reports/
```

Create the following files:

```
cyber_lab/evidence/network/traffic.txt  
cyber_lab/evidence/system/processes.txt  
cyber_lab/logs/security.log  
cyber_lab/scripts/check.sh  
cyber_lab/reports/investigation.txt
```

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

Hint: `tree`

Explore options

Task	Command	Output / Observation	Explanation
Create the directory hierarchy	mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports	No output	`mkdir` creates directories. `-p` creates required parent directories automatically.
Create the required files	touch cyber_lab/evidence/network/traffic.txt cyber_lab/evidence/system/processes.txt cyber_lab/logs/security.log cyber_lab/scripts/check.sh cyber_lab/reports/investigation.txt	No output	`touch` creates empty files with the specified names.
Verify the hierarchy	tree cyber_lab	Displays `cyber_lab` with `evidence`, `logs`, `scripts`, `reports`, and required files.	`tree` displays the directory structure in a tree-like format.
Explore `tree` options	tree --help	Displays available `tree` options.	Shows options used to control the directory display.

```
(sri@kali)-[~]
└─$ mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports

(sri@kali)-[~]
└─$ ls
cyber_lab  Documents  filename.txt  Pictures  Public  scripts  searchlog.sh  Templates  Videos
Desktop    Downloads  Music        Projects  script1.sh  script.sh  system.log    testfiles

(sri@kali)-[~]
└─$ touch cyber_lab/evidence/network/traffic.txt cyber_lab/evidence/system/processes.txt cyber_lab/logs/security.log cyber_lab/scripts/check.sh cyber_lab/reports/investigation.txt

(sri@kali)-[~]
└─$ tree cyber_lab
cyber_lab
├── evidence
│   ├── network
│   │   └── traffic.txt
│   ├── system
│   │   └── processes.txt
│   └── logs
│       └── security.log
├── reports
│   └── investigation.txt
├── scripts
│   └── check.sh
└── 7 directories, 5 files

(sri@kali)-[~]
└─$ tree --help
usage: tree [-acdfghlnpqrstuvxACDFJQNSUX] [-L level [-R]] [-M [-]baseHREF]
        [-I title] [-o filename] [-P pattern] [-I pattern] [--gitignore]
        [--gitfile[=file]] [--matchdirs] [--metafirst] [--ignore-case]
        [--nolinks] [--hintro[=file]] [--houtro[=file]] [--inodes] [--device]
        [--sort[=name]] [--dirsfirst] [--filesfirst] [--filelimit[=#]] [--si]
        [--du] [--prune] [--charset[=X]] [--timefmt[=format]] [--fromfile]
        [--frontabfile] [--fflinks] [--info] [--infofile[=file]] [--noreport]
        [--hyperlink] [--scheme[=schema]] [--authority[=host]] [--opt-toggle]
        [--compress[=#]] [--condense] [--version] [--help] [--acl] [--selinux]
        [--] [directory ...]

Listing options
-a      All files are listed.
-d      List directories only.
-l      Follow symbolic links like directories.
-f      Print the full path prefix for each file.
-x      Stay on current filesystem only.
-L level Descend only level directories deep.
-R      Rerun tree when max dir level reached.
-P pattern List only those files that match the pattern given.
-P pattern Do not list files that match the given pattern.
```

Exercise 8 – Relative and Absolute Paths

Assume the current directory is:

~/cyber_lab

Question	Command	Output / Observation	Explanation
1. Navigate to `evidence/network` using a relative path.	cd evidence/network pwd	~/cyber_lab/evidence/network	The path starts from the current directory, so it is relative.
2. Navigate to `/etc` using an absolute path.	cd /etc pwd	/etc	The path starts from `/`, so it is absolute.
3. Return to your home directory using the shortest possible command.	cd pwd	/home/sri	`cd` without a path directly takes you to your home directory.

Question	Command	Output / Observation	Explanation
4. Navigate to <code>cyber_lab/reports`</code> .	<code>cd ~/cyber_lab/reports</code> <code>pwd</code>	<code>/home/sri/cyber_lab/reports</code>	<code>~`</code> represents your home directory.
5. Move one directory upward.	<code>cd ..</code> <code>pwd</code>	<code>/home/sri/cyber_lab</code>	<code>..`</code> represents the parent directory.
6. Display the current directory after every operation.	<code>pwd</code>	Displays the current absolute path	<code>pwd`</code> shows the exact current location.

Definitions

Absolute path:

A path that starts from the root `/`` and gives the complete location of a file or directory. Example: `/home/sri/cyber_lab/reports``

Relative path:

A path that starts from the current working directory instead of `/``. Example: `evidence/network``

Difference between `/`` and `~``:

- `/`` → Root directory, the starting point of the entire Linux filesystem.
- `~`` → Home directory of the currently logged-in user.

`/`` → Root of Kali Linux
`~`` → `/home/sri``

```
(sri@kali)-[~/cyber_lab]
$ cd evidence/network

(sri@kali)-[~/cyber_lab/evidence/network]
$ pwd
/home/sri/cyber_lab/evidence/network

(sri@kali)-[~/cyber_lab/evidence/network]
$ cd /etc

(sri@kali)-[/etc]
$ pwd
/etc

(sri@kali)-[/etc]
$ cd

(sri@kali)-[~]
$ pwd
/home/sri

(sri@kali)-[~]
$ file cyber_lab/evidence/network/traffic.txt
cyber_lab/evidence/network/traffic.txt: empty

(sri@kali)-[~]
$ ls -l cyber_lab/evidence/network/traffic.txt
-rw-rw-r-- 1 sri sri 0 Sep 13 21:57 cyber_lab/evidence/network/traffic.txt
```


Exercise 9 – Identify File Types

Investigate the following:

```
`/etc/passwd` `/etc` `/bin/bash` `/dev/null` `/proc`
```

Also investigate the files created in Exercise 7.

```
file <filename>
ls -l <filename>
```

Classify each object.

Use the two commands separately for each object:

- `file <filename>` → identifies the type of the object.
- `ls -l <filename>` → shows details and helps confirm the object type from the first character of the permissions.

Object	<code>file` command</code>	<code>ls -l` command</code>	Type
<code>/etc/passwd</code>	file /etc/passwd Output: ASCII text or similar	ls -l /etc/passwd Observe first character <code>-`</code>	Regular file
<code>/etc</code>	file /etc Output: directory	ls -ld /etc Observe first character <code>d`</code>	Directory
<code>/bin/bash</code>	file /bin/bash Output: ELF ... executable	ls -l /bin/bash Observe first character <code>-`</code>	Executable file
<code>/dev/null</code>	file /dev/null Output: character special	ls -l /dev/null Observe first character <code>c`</code>	Character device
<code>/proc</code>	file /proc Output: directory	ls -ld /proc Observe first character <code>d`</code>	Directory / virtual filesystem
<code>security.log</code>	file cyber_lab/logs/security.log Output: empty if created using touch	ls -l cyber_lab/logs/security.log Observe first character <code>-`</code>	Regular file
<code>check.sh</code>	file cyber_lab/scripts/check.sh Output: empty if created using touch	ls -l cyber_lab/scripts/check.sh Observe first character <code>-`</code>	Regular file

Final Classification

Object	Type
<code>/etc/passwd</code>	Regular file
<code>/etc</code>	Directory
<code>/bin/bash</code>	Executable file
<code>/dev/null</code>	Character device
<code>/proc</code>	Directory / virtual filesystem
<code>security.log</code>	Regular file
<code>check.sh</code>	Regular file

Important

For `/etc`` and `/proc``, use:

```
ls -ld /etc
ls -ld /proc
```

instead of `ls -l /etc`` because `ls -l /etc`` shows the contents inside the directory. The `-d`` option shows information about the directory itself.

```
- → Regular file
d → Directory
c → Character device
b → Block device
l → Symbolic link
```

```
-rw-r--r-- → Regular file
drwxr-xr-x → Directory
crw-rw-rw- → Character device
```

```

(sri@kali)-[~]
$ ls -ld /etc
drwxr-xr-x 186 root root 12288 Sep 13 21:33 /etc

(sri@kali)-[~]
$ file /etc
/etc: directory

(sri@kali)-[~]
$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3260 Sep  3 18:54 /etc/passwd

```

Exercise 10 – Examine File Metadata

Use the file:

```
~/linux/theory/t3/cyber_lab/reports/investigation.txt
```

1. Add a small amount of text

```
cd ~/linux/theory/t3/cyber_lab/reports
echo "Security investigation report" > investigation.txt
```

2. Examine the file

Run these separately:

```
ls -l investigation.txt
stat investigation.txt
```

3. Metadata

Item	Command	What to Observe	Answer / Meaning
Filename	stat investigation.txt	File: investigation.txt	investigation.txt
File size	stat investigation.txt	Size: 30	30 bytes
File type	stat investigation.txt	regular file	Regular file
Owner and Group	ls -l investigation.txt	Sri sri	Owner = sri, Group = sri
Inode number	stat investigation.txt	Inode: <number>	Unique number identifying the file
Access time	stat investigation.txt	Access: <date/time>	Last time the file was accessed
Modification time	stat investigation.txt	Modify: <date/time>	Last time the file contents were modified
Change time	stat investigation.txt	Change: <date/time>	Last time the file's metadata/status was changed

Your observed `ls -l` output

```
-rw-rw-r-- 1 sri sri 30 Sep 3 19:18 investigation.txt
```

- Filename: `investigation.txt`
- Size: `30 bytes`
- Owner: `sri`
- Group: `sri`
- Type: Regular file

For the remaining values, use the actual output of:

```
stat investigation.txt
```

Answers

1. What is metadata?

Metadata is information about a file, such as its name, size, owner, permissions, inode number, and timestamps.

2. Is metadata the same as file content?

No. File content is the actual data stored in the file, while metadata describes the file.

3. Which timestamp changes when file contents are modified?

The `Modify` timestamp changes when the contents are modified. The `Change` timestamp normally changes as well because the file's status is updated.

```
(sri@kali)-[~/cyber_lab/reports]
$ echo "security investigation report" > investigation.txt

(sri@kali)-[~/cyber_lab/reports]
$ ls -l investigation.txt
-rw-rw-r-- 1 sri sri 30 Sep 13 22:06 investigation.txt

(sri@kali)-[~/cyber_lab/reports]
$ stat investigation.txt
File: investigation.txt
Size: 30          Blocks: 8          IO Block: 4096   regular file
Device: 8,1      Inode: 739649      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   sri)   Gid: ( 1000/   sri)
Access: 2026-09-13 21:57:42.348596385 +0530
Modify: 2026-09-13 22:06:00.830536414 +0530
Change: 2026-09-13 22:06:00.830536414 +0530
Birth: 2026-09-13 21:57:42.348596385 +0530

(sri@kali)-[~/cyber_lab/reports]
$ ls -l investigation.txt / stat investigation.txt
ls: cannot access 'stat': No such file or directory
-rw-rw-r-- 1 sri sri 30 Sep 13 22:06 investigation.txt
-rw-rw-r-- 1 sri sri 30 Sep 13 22:06 investigation.txt

/:
total 64
lrwxrwxrwx 1 root root 7 Sep 3 17:26 bin -> usr/bin
drwxr-xr-x 3 root root 4096 Sep 3 18:55 boot
drwxr-xr-x 18 root root 3280 Sep 11 23:45 dev
drwxr-xr-x 186 root root 12288 Sep 13 21:33 etc
```

Exercise 11 – Explore `/etc`

These commands only ****read/display**** the files; they do not modify them.

Investigation

Item	Command	What to Observe	Purpose / Information
/etc	ls /etc	passwd, group, hostname, hosts and many other configuration files	Contains system-wide configuration files and directories.
passwd	cat /etc/passwd	Lines containing usernames and account information	Stores user account information such as username, UID, GID, home directory, and login shell.
group	cat /etc/group	Group names and associated group IDs/users	Stores information about groups configured on the system.

Item	Command	What to Observe	Purpose / Information
hostname	cat /etc/hostname	Your Kali system's hostname	Contains the hostname assigned to the system.
hosts	cat /etc/hosts	IP addresses mapped to hostnames	Provides local hostname-to-IP address mappings.

Answers

1. What kind of information is primarily stored under `/etc`?

`/etc` primarily stores system-wide configuration files used by Linux and its applications.

2. What information does `/etc/passwd` contain?

It contains information about user accounts, including usernames, UIDs, GIDs, home directories, and login shells. It does not normally contain the actual user passwords.

3. What information does `/etc/hostname` contain?

It contains the hostname of the Linux system, which is the name used to identify the system on a network.

4. Why is `/etc` security-sensitive?

`/etc` contains important system and security configuration files. Unauthorized modification could change user accounts, authentication, networking, services, or other system behavior.

```
(sri@kali)-[~/cyber_lab/reports]
$ ls /etc
adduser.conf      deluser.conf      hosts              logrotate.conf    openvpn            rmt                sysctl.d
alternatives       depmod.d           hosts.allow        logrotate.d        opt                rpc                sysstat
apache2            dhcpcd.conf        hosts.deny         macchanger         os-release         runit              systemd
apparmor           dhcpd.conf         idmapd.conf        machine-id         pam.conf           samba              terminfo
apparmor.d         dictionaries-common ifplugd            magic             pam.d              sane.d             texmf
apt                doc-base           ImageMagick-7      magic.mime         paperspecs         scalpel            theHarvester
arp-scan           dpkg               inetsim            manpath.config    passwd             screenrc           tightvncserver.con
avahi              e2scrub.conf       init.d             matplotlibrc      passwd-            sddm.conf.d        timidity
bash.bashrc        eac                initscripts         mime.types         perl               searchsploit_rc    tmpfiles.d
bash_completion    emacs              inputrc            minicom            php                security           tpm2-tss
bash_completion.d  environment        ippusbutils        miredo.conf       plymouth            sensors3.conf      ts.conf
bindresvport.blacklist binfmt.d            ipsec.conf         mke2fs.conf       polkit-1            sensors.d           ucf.conf
bluetooth          ettercap            ipsec.d            ModemManager       postgresql           services           udev
ca-certificates    feroxbuster        ipsec.secrets      modprobe.d         postgresql-common   sgml               udisks2
ca-certificates.conf firebird            issue              modules            powershell-empire  shadow             ufw
chatscripts         firefox-esr         issue.net          modules-load.d     profile             shells             unicornscan
chromium            fonts              java-25-openjdk    mosquitto          profile.d            shells             updatedb.conf
chromium.d          freetds            john               mtab               protocols            skel               update-motd.d
cifs-utils          fstab              kali-menu          mysql              proxysql             smartd.conf        UPower
cloud               fuse.conf          kernel             nanorc             pulse               smartmontools      usb_modeswitch.con
colord              gai.conf           kernel-img.conf    netconfig          python2.7            smi.conf           usb_modeswitch.d
console-setup       geoclue            kismet            netsniff-ng        python3              snmp               vconsole.conf
cracklib            ghostscript        ldap              network            python3.13           speech-dispatcher  vdpau_wrapper.cfg
credstore           glvnd              ld.so.cache        NetworkManager     python3.14           sqlmap             vmware-tools
credstore.encrypted gnome-system-tools ld.so.conf          networks            rc0.d               ssl                vpnc
cron.d              gophish            ld.so.conf          networks            rc0.d               ssl                vulkan
```

```
(sri@kali)-[~/cyber_lab/reports]
$ ls /etc | head -n 5
adduser.conf
alternatives
apache2
apparmor
apparmor.d

(sri@kali)-[~/cyber_lab/reports]
$ cat /etc/passwd | head -n 5
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync

(sri@kali)-[~/cyber_lab/reports]
$ cat /var/log | head -n 5
cat: /var/log: Is a directory

(sri@kali)-[~/cyber_lab/reports]
$ ls /var/log | head -n 5
alternatives.log
apache2
apt
boot.log
boot.log.1

(sri@kali)-[~/cyber_lab/reports]
$ lsblk | head -n 5
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 20G 0 disk
sda1 8:1 0 18.9G 0 part /
sda2 8:2 0 1K 0 part
```

Exercise 12 – Explore Linux Log Storage

Investigate `/var/log` and identify:

- Available log files
- Available log directories
- Recently modified entries
- Entries you can read
- Entries you cannot read as a normal user

Commands and Meaning

1. List available log files and directories

```
ls -lah /var/log
```

Meaning: Lists all entries with detailed information and human-readable sizes.

2. Find log files

```
find /var/log -maxdepth 1 -type f
```

Meaning: Finds regular files directly inside `/var/log`.

```
(sri@kali) ~/cyber_lab/reports
$ ls -lah /var/log
total 1.7M
drwxr-xr-x 22 root      root      4.0K Sep 13 19:43 .
drwxr-xr-x 12 root      root      4.0K Sep  5 17:40 ..
-rw-r--r--  1 root      root      85K Sep  3 18:54 alternatives.log
drwxr-x--  2 root      adm       4.0K Sep  3 18:45 apache2
drwxr-xr-x  2 root      root       4.0K Sep 11 23:50 apt
-rw-----  1 root      root        0 Sep 13 19:43 boot.log
-rw-----  1 root      root     14K Sep 13 19:43 boot.log.1
-rw-rw----  1 root      utmp      768 Sep 11 23:45 btmp
-rw-r--r--  1 root      root     1.4M Sep 11 23:50 dpkg.log
-rw-r--r--  1 root      root     7.2K Sep  3 18:49 fontconfig.log
drwxr-xr-x  2 _gvm      _gvm      4.0K Apr 28 19:26 gvm
drwx-----  3 inetsim   inetsim    4.0K Sep  3 18:45 inetsim
drwxr-xr-x  3 root      root       4.0K Sep  3 20:16 installer
drwxr-sr-x+  3 root      systemd-journal 4.0K Sep  5 17:40 journal
-rw-rw-r--  1 root      utmp        0 Sep  3 17:26 lastlog
drwx--x--x  2 root      root       4.0K Sep 11 23:45 lightdm
-rw-r--r--  1 root      root        59 Sep 11 23:45 macchanger.log.1.gz
drwxr-----  2 mosquito mosquito    4.0K May 14 15:13 mosquito
drwxr-s--  2 mysql      adm       4.0K Sep  3 18:44 mysql
drwxr-xr-x  2 root      adm       4.0K Sep  3 18:39 nginx
drwxr-xr-x  2 _gvm      _gvm      4.0K Apr 16 11:12 notus-scanner
drwxr-xr-x  2 root      root       4.0K Apr 28 02:55 openvpn
drwxrwxr-t  2 root      postgres   4.0K Sep  3 18:45 postgresql
drwx-----  2 root      root       4.0K Sep  3 17:26 private
drwxr-s--  2 redis      adm       4.0K Sep  3 18:40 redis
drwxr-xr-x  4 root      root       4.0K Sep  3 18:38 runit
drwxr-x--  2 root      adm       4.0K May 26 18:16 samba
drwx-----  2 speech-dispatcher root      4.0K Sep  3 18:45 speech-dispatcher
drwxr-xr-x  2 stunnel    stunnel    4.0K Sep  3 18:41 stunnel
drwxr-xr-x  2 root      root       4.0K Jan 16  2024 sysstat
-rw-----  1 root      root      217 Sep  5 17:40 vmware-network.1.log
-rw-----  1 root      root      213 Sep 11 23:45 vmware-network.log
-rw-----  1 root      root     2.9K Sep  5 12:16 vmware-vmtoolsd-root.1.log
-rw-----  1 root      root     2.6K Sep 13 19:43 vmware-vmtoolsd-root.log
-rw-----  1 root      root     1.1K Sep 11 23:45 vmware-vmtoolsd-sri.log
-rw-----  1 root      sri       1.2K Sep 11 23:46 vmware-vmtoolsd-sri.log
-rw-----  1 root      sri       3.1K Sep 11 23:46 vmware-vmtoolsd-sri.log
-rw-rw-r--  1 root      utmp      1.9K Sep 11 23:46 wtmp
-rw-r--r--  1 root      root      32K Sep 13 20:08 Xorg.0.log
-rw-r--r--  1 root      root      32K Sep  5 12:16 Xorg.0.log.old

(sri@kali) ~/cyber_lab/reports
$ find /var/log -maxdepth 1 -type f
/var/log/vmware-network.1.log
/var/log/wtmp
/var/log/macchanger.log.1.gz
/var/log/vmware-vmtoolsd-root.log
/var/log/fontconfig.log
/var/log/vmware-vmtoolsd-sri.log
/var/log/btmp
/var/log/vmware-vmtoolsd-root.log
```



```

(sri@kali)-[~/cyber_lab/reports]
$ ls -lah /tmp | head -n 4
total 12K
drwxrwxrwt 13 root root 320 Sep 13 22:09 .
drwxr-xr-x 19 root root 4.0K Sep  3 18:46 ..
-rw----- 1 sri sri 0 Sep 11 23:46 config-err-58iu3n

(sri@kali)-[~/cyber_lab/reports]
$ ls -la /tmp | head -n 4
total 12
drwxrwxrwt 13 root root 320 Sep 13 22:09 .
drwxr-xr-x 19 root root 4096 Sep  3 18:46 ..
-rw----- 1 sri sri 0 Sep 11 23:46 config-err-58iu3n

(sri@kali)-[~/cyber_lab/reports]
$ ls -l /tmp | head -n 4
total 0
-rw----- 1 sri sri 0 Sep 11 23:46 config-err-58iu3n
drwx----- 3 root root 60 Sep 11 23:46 systemd-private-2a8a1b808c39440aa40b71be376d3a4f-colord.service-fRmhfo
drwx----- 3 root root 60 Sep 11 23:45 systemd-private-2a8a1b808c39440aa40b71be376d3a4f-ModemManager.service-861YzL

(sri@kali)-[~/cyber_lab/reports]
$ ls -lt /tmp | head -n 4
total 0
drwx----- 3 root root 60 Sep 11 23:46 systemd-private-2a8a1b808c39440aa40b71be376d3a4f-upower.service-Ppsa7Y
-rw----- 1 sri sri 0 Sep 11 23:46 config-err-58iu3n
drwx----- 3 root root 60 Sep 11 23:46 systemd-private-2a8a1b808c39440aa40b71be376d3a4f-colord.service-fRmhfo

```

3. Find log directories

```
find /var/log -maxdepth 1 -type d
```

Meaning: Finds directories directly inside `/var/log``.

4. Find recently modified entries

```
ls -lt /var/log
```

Meaning: Lists entries with the newest modified entries first.

5. Find readable entries

```
find /var/log -maxdepth 1 -readable -print
```

Meaning: Finds entries readable by the current user.

6. Find non-readable entries

```
find /var/log -maxdepth 1 ! -readable -print
```

Meaning: Finds entries that are not readable by the current user.

Meaning of Options

Option	Meaning	Option	Meaning
<code>-l</code>	Long/detailed listing	<code>-a</code>	Show all entries
<code>-h</code>	Human-readable sizes	<code>-maxdepth 1</code>	Search only specified directory
<code>-type f</code>	Find files	<code>-type d</code>	Find directories
<code>-t</code>	Newest modified first	<code>-readable</code>	Find readable entries
<code>!</code>	NOT condition	<code>-print</code>	Print matching entries

Why No ``sudo``?

Do not use ``sudo`` when checking readable and non-readable entries because the exercise asks what a **normal user** can access. ``sudo`` gives administrator privileges and can change the result.

Answers

1. Why does Linux maintain logs?

Linux maintains logs to record system, application, and security events. These records help track what happened on the system.

2. Why are logs useful during incident investigation?

Logs provide evidence of activities such as logins, errors, and suspicious events. They help investigators reconstruct a timeline of an incident.

3. Why might access to some logs be restricted?

Some logs contain sensitive system or security information. Restricting access prevents unauthorized users from viewing or modifying them.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

- Whether files/directories currently exist.
- Whether hidden files exist.
- Who owns some of the entries.
- Which entries were modified recently.
-

Commands and Meaning

1. List files and directories

```
ls -lah /tmp | head -n 4
```

Meaning: Lists `/tmp` entries with details, hidden entries, and human-readable sizes.

2. Check hidden files

```
ls -la /tmp | head -n 4
```

Meaning: Lists all entries, including hidden entries.

```
(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev | head -n 5
total 0
crw-r--r--  1 root root    10, 235 Sep 11 23:45 autofs
drwxr-xr-x  2 root root    140 Sep 11 23:46 block
drwxr-xr-x  2 root root     80 Sep 11 23:45 bsg
crw-----  1 root root    10, 234 Sep 11 23:45 btrfs-control

(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev/null | head -n 4
crw-rw-rw-  1 root root  1, 3 Sep 11 23:45 /dev/null

(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev/zero
crw-rw-rw-  1 root root  1, 5 Sep 11 23:45 /dev/zero

(sri@kali)-[~/cyber_lab/reports]
$ file /dev/null
/dev/null: character special (1/3)

(sri@kali)-[~/cyber_lab/reports]
$ file /dev/zero
/dev/zero: character special (1/5)
```

3. Check ownership

```
ls -l /tmp | head -n 4
```

Meaning: Shows permissions, owner, group, size, and modification time.

4. Find recently modified entries

```
ls -lt /tmp | head -n 4
```

Meaning: Lists entries with the newest modified entries first.

Meaning of Options

Option	Meaning	Option	Meaning
-l	Long/detailed listing	-a	Show all entries
-h	Human-readable sizes	-t	Newest modified first
head -n 4	Show first 4 lines	/tmp	Directory being investigated

Observation Table

Investigation	Result from Your Output
Files/directories exist	Yes, files and directories are present
Hidden entries	. and .. are present
Ownership	config-err-Ncf65G → sri:sri; systemd-private-* → root:root
Recently modified	systemd-private-...systemd-logind... → 23:23, the newest shown

How to Identify Ownership

```
-rw----- 1 sri sri 0 Sep 4 22:35 config-err-Ncf65G
`sri` = **owner**
second `sri` = **group**

drwx----- 3 root root 60 Sep 4 23:23 systemd-private-...
`root` = **owner**
second `root` = **group**
```

Answer

Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories may contain temporary files, scripts, downloads, or other artifacts created during suspicious activity. They can provide useful evidence about recent system or user activity.

Exercise 14 – Explore the `/proc` Filesystem

Investigate: `/proc`

Examine:

```
cat /proc/cpuinfo
cat /proc/meminfo
```

Then identify your current shell PID:

```
echo $$
```

Check whether a corresponding directory exists under `/proc`.

```
(sri@kali)-[~/cyber_lab/reports]
$ cat /proc/cpuinfo | head -n 4
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 142

(sri@kali)-[~/cyber_lab/reports]
$ cat /proc/meminfo | head -n 5
MemTotal:      1979692 kB
MemFree:       277324 kB
MemAvailable:  742712 kB
Buffers:       38716 kB
Cached:        530972 kB

(sri@kali)-[~/cyber_lab/reports]
$ echo $$
3591

(sri@kali)-[~/cyber_lab/reports]
$ ls -ld /proc/$$
dr-xr-xr-x 9 sri sri 0 Sep 11 23:48 /proc/3591
```

Commands and Meaning

Task	Command	Output / Observation	Meaning
Examine CPU information	cat /proc/cpuinfo head -n 4	processor : 0; vendor_id : AuthenticAMD; cpu family : 23; model : 160	Displays CPU information such as processor number, vendor, family, and model.
Examine memory information	cat /proc/meminfo head -n 5	MemTotal: 2016604 kB; MemFree: 299724 kB; MemAvailable: 1209664 kB; Buffers: 89740 kB; Cached: 919504 kB	Displays system memory statistics.
Identify current shell PID	echo \$\$	2030	2030 is the PID of the current shell.
Check the PID directory	ls -ld /proc/\$\$	dr-xr-xr-x 9 sri sri 0 Sep 4 22:39 /proc/2030	Confirms that a directory for the current shell process exists under <code>/proc`</code> .

Answers

1. What information does `/proc/cpuinfo`` provide?

It provides CPU details such as the processor number, vendor, CPU family, and model.

2. What information does `/proc/meminfo`` provide?

It provides memory statistics such as total, free, available, buffered, and cached memory.

3. Why do directories with numbers appear inside `/proc``?

They represent running processes. Each number is a process ID (PID).

4. Is `/proc`` an ordinary disk directory?

No. `/proc`` is a virtual filesystem created by the Linux kernel.

5. Why could `/proc`` be useful during security investigation?

It provides information about running processes and system resources, which can help identify suspicious activity.

Exercise 15 – Explore `/dev``

Investigate `/dev`` and examine `/dev/null`` and `/dev/zero``.

Commands and Meaning

1. Examine `/dev/null``

```
ls -l /dev/null
```

Meaning: Displays the permissions, owner, group, and type of `/dev/null``.

2. Examine `/dev/zero``

```
ls -l /dev/zero
```

Meaning: Displays the permissions, owner, group, and type of `/dev/zero``.

3. Determine the file type

```
file /dev/null
```

Meaning: Identifies the type of `/dev/null``.

Task	Command	Output / Observation	Meaning
Check <code>/dev`</code>	/dev	bash: /dev: Is a directory	<code>/dev`</code> is a directory and cannot be executed as a command.
Check <code>/dev/null`</code>	/dev/null	bash: /dev/null: Permission denied	<code>/dev/null`</code> is a special device file, not a normal executable file.
Examine <code>/dev/null`</code>	ls -l /dev/null	crw-rw-rw- 1 root root 1, 3 Sep 4 22:34 /dev/null	First character <code>c`</code> shows a character device.
Examine <code>/dev/zero`</code>	ls -l /dev/zero	crw-rw-rw- 1 root root 1, 5 Sep 4 22:34 /dev/zero	First character <code>c`</code> shows a character device.

Meaning of Options

Option	Meaning	Option	Meaning
-l	Long/detailed listing	/dev	Device filesystem
c	Character device	file	Identifies file type

Answers

1. What does `/dev` represent?

/dev contains special files representing hardware and virtual devices used by Linux.

2. Are the entries ordinary user files?

No. They are special device files used by the operating system to communicate with devices.

3. Determine the file type of `/dev/null`.

/dev/null is a **character device**, indicated by `c` at the beginning of the `ls -l` output.

4. Why does Linux expose devices through the filesystem?

It provides a standard way for programs and users to interact with hardware and virtual devices using filesystem operations.

```
(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev | head -n 5
total 0
crw-r--r--  1 root root    10, 235 Sep 11 23:45 autofs
drwxr-xr-x  2 root root    140 Sep 11 23:46 block
drwxr-xr-x  2 root root     80 Sep 11 23:45 bsg
crw-----  1 root root    10, 234 Sep 11 23:45 btrfs-control

(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev/null | head -n 4
crw-rw-rw-  1 root root  1, 3 Sep 11 23:45 /dev/null

(sri@kali)-[~/cyber_lab/reports]
$ ls -l /dev/zero
crw-rw-rw-  1 root root  1, 5 Sep 11 23:45 /dev/zero

(sri@kali)-[~/cyber_lab/reports]
$ file /dev/null
/dev/null: character special (1/3)

(sri@kali)-[~/cyber_lab/reports]
$ file /dev/zero
/dev/zero: character special (1/5)
```

Exercise 16 – Filesystem Space Investigation

Determine:

- Total filesystem capacity
- Used space
- Available space
- Percentage of filesystem utilization
- Space occupied by your home directory
- Space occupied by the `cyber\_lab` directory

Use appropriate combinations of:

```
df
df -h
du
du -h
du -sh
```

Commands and Meaning

Task	Command	Meaning
Check filesystem space	df	Shows filesystem capacity, used space, available space, and utilization.
Human-readable filesystem space	df -h	Displays filesystem sizes in readable units such as GB and MB.
Check directory space	du	Shows disk space used by files and directories.
Human-readable directory space	du -h	Displays directory usage in readable units.
Check total directory size	du -sh	Shows only the total size of the specified directory.

Commands for Required Information

1. Check filesystem space

```
df -h
```

Meaning: Shows total, used, available space, and usage percentage in human-readable format.

Use the row containing `/` to record the filesystem containing your root directory.

2. Check home directory space

```
du -sh "$HOME"
```

Meaning: Shows the total space occupied by your home directory.

3. Check `cyber\_lab` directory space

```
du -sh ~/linux/theory/t3/cyber_lab
```

Meaning: Shows the total space occupied by the `cyber\_lab` directory.

```
(sri@kali)-[~/cyber_lab/reports]
$ du -h
8.0K .

(sri@kali)-[~/cyber_lab/reports]
$ du -sh
8.0K .

(sri@kali)-[~/cyber_lab/reports]
$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
udev             954372         0   954372    0% /dev
tmpfs            197972      1196   196776    1% /run
/dev/sda1       19353424 15262800  3082192   84% /
tmpfs            989844         4   989840    1% /dev/shm
none             1024         0     1024    0% /run/credentials/systemd-journald.service
tmpfs            989848      756   989092    1% /tmp
none             1024         0     1024    0% /run/credentials/getty@tty1.service
tmpfs            197968      116   197852    1% /run/user/1000

(sri@kali)-[~/cyber_lab/reports]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            933M   0  933M   0% /dev
tmpfs           194M  1.2M  193M   1% /run
/dev/sda1       19G   15G   3.0G  84% /
tmpfs           967M  4.0K  967M   1% /dev/shm
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M  768K  966M   1% /tmp
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  116K  194M   1% /run/user/1000

(sri@kali)-[~/cyber_lab/reports]
$ du
8

(sri@kali)-[~/cyber_lab/reports]
$ du -h
8.0K .

(sri@kali)-[~/cyber_lab/reports]
$ du -sh
8.0K .
```


Observation

Item	Result
Total filesystem capacity	From `df -h`
Used space	From `df -h`
Available space	From `df -h`
Filesystem utilization	From `df -h`
Home directory size	From `du -sh "\$HOME"`
`cyber_lab` size	From `du -sh ~/linux/theory/t3/cyber_lab`

Answer

`df` shows space available and used on the **filesystem as a whole**, while `du` shows space used by **individual files and directories**.

```
(sri@kali)~[/cyber_lab/reports]
$ du -sh "$HOME"
2.2M    /home/sri

(sri@kali)~[/cyber_lab/reports]
$ cd cyber_lab
cd: no such file or directory: cyber_lab

(sri@kali)~[/cyber_lab/reports]
$ ls
investigation.txt

(sri@kali)~[/cyber_lab/reports]
$ cd ..

(sri@kali)~[/cyber_lab]
$ du
8       ./reports
4       ./logs
4       ./scripts
4       ./evidence/system
4       ./evidence/network
12      ./evidence
32      .

(sri@kali)~[/cyber_lab]
$ cd ..

(sri@kali)~[~]
$ du
4       ./Documents
4       ./Public
12      ./scripts/testfiles
36      ./scripts
4       ./Templates
8       ./config/pulse
4       ./config/enchant
8       ./config/dconf
72      ./config/xfce4/xfconf/xfce-perchannel-xml
76      ./config/xfce4/xfconf
8       ./config/xfce4/panel/launcher-5
16      ./config/xfce4/panel/launcher-7
```

```

(sri@kali)-[~]
$ cd cyber_lab

(sri@kali)-[~/cyber_lab]
$ du
8      ./reports
4      ./logs
4      ./scripts
4      ./evidence/system
4      ./evidence/network
12     ./evidence
32     .

(sri@kali)-[~/cyber_lab]
$ du -h
8.0K   ./reports
4.0K   ./logs
4.0K   ./scripts
4.0K   ./evidence/system
4.0K   ./evidence/network
12K    ./evidence
32K    .

(sri@kali)-[~/cyber_lab]
$ du -sh
32K    .

(sri@kali)-[~/cyber_lab]
$ du -sh "$HOME"
2.2M   /home/sri

(sri@kali)-[~/cyber_lab]
$ cd ..

```

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported. Investigate the following filesystem locations and identify why each is important.

Investigation Table

Location	Why Investigate?	Security-Relevant Information
/home	Contains files and directories belonging to normal users.	Suspicious files, scripts, downloads, hidden files, and user activity.
/etc	Contains important system and service configuration files.	User accounts, authentication, networking, services, and system configuration.
/var/log	Contains system and application logs.	Login activity, errors, service events, and suspicious activity.
/tmp	Contains temporary files created by users and applications.	Temporary scripts, downloads, files, and possible attack artifacts.
/proc	Provides information about the running system and processes.	Running processes, PIDs, memory information, and current system activity.
/dev	Contains special files representing devices.	Device information and unusual or suspicious device entries.
/root	Contains files belonging to the root user.	Root-user files, scripts, configuration files, and possible suspicious activity.

Conclusion

All these locations are relevant during a filesystem-based cybersecurity investigation. Together, they can provide evidence about **users, configurations, logs, temporary artifacts, running processes, devices, and root-user activity**.

Submit the following

- Commands used for each exercise.
- Relevant output or screenshots.
- Observation tables.
- Disk → Partition → Filesystem → Mount Point diagram.
- Answers to analysis questions.

****Important:**** Do not modify or delete system files while performing the exercises.

Linux Filesystem – Command Cheat Sheet

Exercise	Commands
1. Filesystem	lsblk → devices/partitions lsblk -f → filesystem type df -T → filesystem & space df -h → readable space
2. Storage Diagram	Use output from Exercise 1 → Disk → Partition → Filesystem → Mount Point
3. Root Filesystem	pwd → current path cd / → root ls → contents ls -la → all entries
4. Hierarchy	ls -lah /home ls -lah /root ls -lah /etc ls -lah /var ls -lah /var/log ls -lah /tmp ls -lah /usr ls -lah /boot ls -lah /dev ls -lah /proc
5. Home Directory	whoami → username echo \$HOME → home path cd "\$HOME" → home pwd → absolute path ls → normal contents ls -a → all contents
6. Hidden Files	ls → normal entries ls -a → hidden entries
7. Build Hierarchy	mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports touch cyber_lab/evidence/network/traffic.txt touch cyber_lab/evidence/system/processes.txt touch cyber_lab/logs/security.log touch cyber_lab/scripts/check.sh touch cyber_lab/reports/investigation.txt tree cyber_lab
8. Paths	cd evidence/network → relative cd /etc → absolute cd ~ → home cd ~/cyber_lab/reports cd .. → one level up pwd → current path
9. File Types	file <filename> → type ls -l <filename> → details
10. Metadata	echo "Investigation report" > investigation.txt ls -l investigation.txt stat investigation.txt
11. /etc	ls -lah /etc cat /etc/passwd cat /etc/group cat /etc/hostname cat /etc/hosts
12. /var/log	ls -lah /var/log find /var/log -maxdepth 1 -type f find /var/log -maxdepth 1 -type d ls -lt /var/log find /var/log -maxdepth 1 -readable -print find /var/log -maxdepth 1 ! -readable -print

Exercise	Commands
13. /tmp	ls -lah /tmp ls -la /tmp ls -l /tmp ls -lt /tmp
14. /proc	cat /proc/cpuinfo head -n 4 cat /proc/meminfo head -n 5 echo \$\$ ls -ld /proc/\$\$
15. /dev	ls -l /dev/null ls -l /dev/zero file /dev/null
16. Space	df df -h du du -h du -sh "\$HOME" du -sh ~/linux/theory/t3/cyber_lab
17. Investigation	ls -lah /home ls -lah /etc ls -lah /var/log ls -lah /tmp ls -lah /proc ls -lah /dev ls -lah /root

☆ Quick Option Meanings

Option	Meaning	Option	Meaning
-l	Detailed listing	-a	All entries
-h	Human-readable	-t	Newest first
-f	Filesystem information	-T	Filesystem type
-s	Summarize size	-d	Directory itself
-p	Create parent directories	-maxdepth 1	Only specified directory
-type f	Files	-type d	Directories
-readable	Readable entries	!	NOT condition
-print	Print results	\$\$	Current shell PID